



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For



CompTIA Certified Linux+ Training (XK0-006)

TGS Ref No: TGS-2024048316

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version v2

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
v1	2 Jul 2026	Initial release — CompTIA Linux+ XK0-006 V8 2-day lesson plan aligned to the 30 labs, the five exam domains and the slide deck (slide numbers cited).	Dr. Alfred Ang
v2	18 Aug 2026	Full teaching chapters from the legacy Learner Guide deck merged into the slide deck and re-ordered onto the five XK0-006 exam domains; exam-domain blueprint slide added at the start; CompTIA exam registration and practice-exam slides added at the end; labs restructured one-folder-per-lab; slide references updated to the v2 deck.	Dr. Alfred Ang

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Course Information

Course Title	CompTIA Certified Linux+ Training (XK0-006)
WSQ Course Reference	TGS-2024048316
Certification Exam	CompTIA Linux+ XK0-006 V8
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 8 training hours per day (16 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on labs on the free KillerCoda Ubuntu Playground
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- Explain core Linux concepts — the boot process, the Filesystem Hierarchy Standard, device management and virtualization — and manage storage, networking and shell operations.
- Manage files, local user and group accounts, processes and jobs, software packages, systemd services and containerized applications on a Linux server.
- Apply Linux security: authentication/authorization (sudo, PAM), firewalls, OS and account hardening, cryptography and compliance/audit procedures.
- Automate administration with Ansible, Bash and Python, apply Git version control, and use AI assistants responsibly and securely.
- Monitor a Linux system and analyze and troubleshoot hardware, storage, network, security and performance issues using the right diagnostic tools.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book — aligned to the course slides.
- Practical Performance (PP) — hands-on Linux tasks, 1 hour, open book — aligned to the course labs.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- The final assessment is conducted on Day 2 from 4:00 pm (WA 4:00–5:00, PP 5:00–6:00).
- A minimum of 75% attendance is required to be eligible for assessment and funding.
- Courseware and the assessment are on the LMS — <https://lms-tms.tertiaryinfotech.com/>.

Exam Domain Coverage

The course maps 1:1 to the CompTIA Linux+ XK0-006 exam blueprint. Each domain, its exam weighting, the labs that cover it, and the deck slides:

Domain	Exam Weight	Labs	Deck Slides
1. System Management	23%	1–7	23–343
2. Services and User Management	20%	8–13	344–879
3. Security	18%	14–19	880–1258
4. Automation, Orchestration, and Scripting	17%	20–24	1259–1655
5. Troubleshooting	22%	25–30	1656–2181

Daily Schedule

Each training day runs 9:30 am – 6:30 pm and delivers 8 instructional hours (1-hour lunch; tea breaks counted within training time). The Slides column cites the deck page range the trainer runs for each teaching session.

Day 1 — System Management & Services / User Management

Time	Duration	Topic / Activity	Method	Slides
9:30–9:45	15 min	Welcome, mandatory digital attendance (AM), trainer & learner introduction	Admin	—
9:45–10:00	15 min	Course overview, learning outcomes and ground rules	Admin	—
10:00–12:15	135 min	Domain 1 — System Management (23%): boot & FHS, kernel modules, LVM storage, networking, shell operations. Hands-on Labs 1–5	Lecture + Hands-on Lab	23–323
12:15–1:15	60 min	Lunch break	Break	—
1:15–2:45	90 min	Domain 1 (cont.): backup & restore, virtualization. Hands-on Labs 6–7	Lecture + Hands-on Lab	324–343
2:45–2:55	10 min	Digital attendance (PM)	Admin	—
2:55–5:45	170 min	Domain 2 — Services and User Management (20%): files, accounts, processes, packages,	Lecture + Hands-on Lab	344–879

		systemd, containers. Hands-on Labs 8–13		
5:45–6:15	30 min	Domain 1–2 recap, Q&A and consolidation	Discussion	—
6:15–6:30	15 min	Day 1 review, preview of Day 2 and PM digital attendance	Discussion	—

Total training time: 480 minutes (8 hours).

Day 2 — Security, Automation, Troubleshooting & Final Assessment

Time	Duration	Topic / Activity	Method	Slides
9:30–9:40	10 min	Mandatory digital attendance (AM) and Day 1 recap	Admin	—
9:40–12:15	155 min	Domain 3 — Security (18%): AAA/sudo/PAM, firewalls, OS & account hardening, cryptography, compliance & audit. Hands-on Labs 14–19	Lecture + Hands-on Lab	880–1258
12:15–1:15	60 min	Lunch break	Break	—
1:15–3:05	110 min	Domain 4 — Automation, Orchestration & Scripting (17%): Ansible, Bash, Python, Git, responsible AI. Hands-on Labs 20–24	Lecture + Hands-on Lab	1259–1655
3:05–3:15	10 min	Digital attendance (PM)	Admin	—
3:15–4:05	50 min	Domain 5 — Troubleshooting (22%): monitoring, storage, network, security, performance. Hands-on Labs 25–29	Lecture + Hands-on Lab	1656–2161
4:05–4:25	20 min	Capstone — end-to-end server build & triage. Hands-on Lab 30	Lecture + Hands-on Lab	2162–2181
4:25–4:30	5 min	Revision · Briefing for Assessment · Course feedback &	Assessment	—

		TRAQOM survey		
4:30–5:30	60 min	Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book	Assessment	—
5:30–6:30	60 min	Practical Performance (PP) — hands-on Linux tasks, 1 hour, open book. PM digital attendance; submit answers on the LMS	Assessment	—

Total training time: 480 minutes (8 hours).

Topic-by-Topic Breakdown

Each lab below cites the deck slide of its overview so the trainer can jump straight to it.

Domain 1 — System Management (23%) • Slides 23–343

Lab 1 — Boot Process & Filesystem Hierarchy [Obj 1.1] • Slide 281

Inspect a running Linux system's boot configuration, kernel command line, initramfs, and the standard Filesystem Hierarchy to understand how GRUB hands off to the kernel and where FHS directories live.

Lab 2 — Kernel Modules & Device Management [Obj 1.2] • Slide 289

List, inspect, load, and unload kernel modules and use the standard hardware-enumeration tools to understand modprobe vs insmod, module dependencies, and reading dmesg.

Lab 3 — Storage with LVM, Partitions & Filesystems [Obj 1.3] • Slide 298

Build a complete LVM stack on loopback-backed disks, format with ext4/xfs/btrfs, mount with hardening options, persist in fstab, and grow a filesystem online.

Lab 4 — Network Configuration [Obj 1.4] • Slide 306

Inspect interfaces, addresses, and routes with ip; walk the name-resolution stack; validate a netplan YAML; and use the standard diagnostic toolbox to read a Linux network stack top-to-bottom.

Lab 5 — Shell Operations & Text Processing [Obj 1.5] • Slide 315

Work with shell environment variables, login vs interactive startup files, every common redirection form, and the classic text-processing pipeline to assemble non-trivial one-liners.

Lab 6 — Backup & Restore [Obj 1.6] • Slide 324

Create tar/cpio/dd archives with gzip/bzip2/xz compression, image and recover disks with dd and ddrescue, mirror trees with rsync, and verify integrity with sha256sum.

Lab 7 — Virtualization with QEMU/KVM & libvirt [Obj 1.7] · Slide 333

Install QEMU and libvirt, manipulate virtual disk images with qemu-img, list libvirt domains/networks with virsh, and boot a tiny guest under QEMU to discuss hypervisors, image formats, snapshots, and network modes.

Domain 2 — Services and User Management (20%) · Slides 344–879

Lab 8 — File & Directory Management [Obj 2.1] · Slide 824

Practice core file and directory operations: navigating, creating, copying, searching, linking, and inspecting special device files with standard Linux utilities.

Lab 9 — Local Account & Group Management [Obj 2.2] · Slide 833

Create and manage local users and groups, set passwords and aging policies, change shells, and distinguish user, system, and service accounts.

Lab 10 — Processes, Jobs & Scheduling [Obj 2.3] · Slide 842

Inspect processes, manipulate shell jobs, adjust priorities, send signals, and schedule tasks with at, cron, and anacron.

Lab 11 — Software & Package Management [Obj 2.4] · Slide 851

Manage Debian packages with apt/dpkg, add a signed third-party repo, run dnf/rpm in a container, and use language and sandboxed package managers.

Lab 12 — Service Management with systemd [Obj 2.5] · Slide 860

Inspect and control systemd units, author custom service/timer/mount units, analyze boot performance, and configure host/time/DNS via systemd helpers.

Lab 13 — Containers with Docker & Podman [Obj 2.6] · Slide 869

Install Docker and Podman, pull/run images, manage volumes and networks, build an image from a Dockerfile, and contrast rootful vs rootless engines.

Domain 3 — Security (18%) · Slides 880–1258

Lab 14 — AAA: sudo, PAM, and Polkit [Obj 3.1] · Slide 1203

Configure sudo authorization, inspect PAM authentication stacks, and enable auditd accounting to record security events. Finish with a conceptual look at centralized identity via LDAP, Kerberos, and SSSD.

Lab 15 — Firewalls: ufw, firewalld, iptables, nftables, ipset [Obj 3.2] · Slide 1212

Configure firewalls at every layer of the Linux netfilter stack, from the ufw wrapper down to iptables and nftables, and understand stateful inspection. Explore zones, rich rules, ipset, and SNAT/DNAT.

Lab 16 — OS Hardening: Permissions, SELinux, SSH, Fail2ban [Obj 3.3] · Slide 1221

Apply layered OS hardening from file permissions and immutability through SELinux labeling, then harden OpenSSH and add brute-force protection. Inventory SUID binaries and legacy cleartext services.

Lab 17 — Account Hardening [Obj 3.4] · Slide 1230

Harden user accounts through password quality and aging policies, brute-force lockouts, and restricted or non-interactive shells. Explore TOTP MFA and Have I Been Pwned k-anonymity lookups.

Lab 18 — Cryptography: GPG, LUKS, OpenSSL, WireGuard [Obj 3.5] · Slide 1239

Practice core Linux cryptography: GPG key generation and signing, LUKS2 disk encryption, OpenSSL certificates, hashing and HMAC, and WireGuard keys. Learn to identify weak TLS versions and ciphers.

Lab 19 — Compliance, Auditing, and File Integrity [Obj 3.6] · Slide 1248

Establish file-integrity monitoring and run compliance and audit tooling to detect tampering, rootkits, and misconfigurations. Cover secure deletion, login banners, and OpenSCAP/CIS benchmarks.

Domain 4 — Automation, Orchestration, and Scripting (17%) · Slides 1259–1655

Lab 20 — Infrastructure as Code with Ansible [Obj 4.1] · Slide 1610

Learn Infrastructure as Code using Ansible as the primary agentless configuration management tool, and recognize the wider IaC ecosystem (Puppet, OpenTofu, cloud-init, Kubernetes ConfigMaps). Build an idempotent playbook that deploys and configures nginx.

Lab 21 — Bash Scripting [Obj 4.2] · Slide 1618

Practice core Bash scripting constructs (variables, conditionals, loops, functions, option parsing, regex, file tests) and assemble a defensive backup script, then audit it with shellcheck.

Lab 22 — Python for System Administration [Obj 4.3] · Slide 1627

Use Python for common Linux admin tasks: create a virtual environment, exercise core data types, walk the filesystem, parse JSON, call a REST API with requests, and enforce PEP 8 with black and flake8.

Lab 23 — Git Version Control [Obj 4.4] · Slide 1636

Work through Git end to end: init, commit, branch, merge, resolve conflicts, stash, tag, rewrite history with reset and rebase, and push/pull against a local bare remote.

Lab 24 — Responsible AI Use for Linux Administrators [Obj 4.5] · Slide 1645

Adopt a verify-before-paste workflow for AI-generated Linux commands: redact secrets before sending prompts to hosted models, lint and sandbox suggestions, weigh local vs hosted models, and draft a corporate AI usage policy.

Domain 5 — Troubleshooting (22%) · Slides 1656–2181

Lab 25 — Monitoring Concepts and Tools [Obj 5.1] · Slide 2117

Learn the monitoring vocabulary (SLA/SLO/SLI) and stand up a working monitoring stack to produce metrics, alerts, webhooks, health checks, and aggregated logs.

Lab 26 — Troubleshooting Hardware, Storage, and OS Issues [Obj 5.2] · Slide 2126

Deliberately inject common OS and storage failures then diagnose and repair each, keeping every breakage contained to loopback devices, throwaway units, or subshells.

Lab 27 — Troubleshooting Network Connectivity [Obj 5.3] · Slide 2135

Break and repair the common network faults (DNS, routing, firewall, MTU, IP, link, dual-stack) while practicing the standard L3/L4 triage toolkit.

Lab 28 — Troubleshooting Security Issues [Obj 5.4] · Slide 2144

Diagnose and repair security faults spanning SELinux contexts, POSIX ACLs, certificate trust, account lockout, broken repos, and weak TLS, using a Rocky container for the SELinux portion.

Lab 29 — Troubleshooting Performance Issues [Obj 5.5] · Slide 2153

Stress CPU, memory, disk I/O, and network in turn, then walk the universal performance loop: baseline, spike, identify the bottleneck with the right counter, and resolve it.

Lab 30 — Capstone: End-to-End Server Build and Triage [Obj Capstone] · Slide 2162

Integrate Objectives 1-5 into one coherent build: a hardened admin account, LVM storage, nginx with a systemd override, a locked firewall, a health-check timer, and git-versioned config, then inject a fault and triage it end-to-end.

Resources Required

- A laptop with a modern web browser and reliable internet access.
- Free KillerCoda Ubuntu Playground — <https://killercoda.com/playgrounds/scenario/ubuntu> (no local install required).
- Course slide deck and Learner Guide (downloaded from the LMS).
- Tertiary Infotech LMS account — <https://lms-tms.tertiaryinfotech.com/> (courseware download and assessment).
- CompTIA Linux+ XK0-006 V8 Exam Objectives (reference blueprint).